Gage William Boyd

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Education

Northern Arizona University — **Expected May 2028**
B.S. Mathematics & B.S. Computer Science | GPA: 3.6/4.0 — *Flagstaff, AZ*

- **Competitions:** Jump Trading Probability Cup (2026) — **top 220 of 4,012** competitors (top 5.5%)
- **Honors & Activities:** Dean's List, Lumberjack Scholarship, SCLA Honors Society, ACM, FAMUS
- **Coursework:** Regression Analysis, Probability & Statistics, Differential Equations, Linear Algebra (2027)

Experience

Machine Learning Research Assistant — **May 2026 – Present**
NAU UQLID Lab — Scientific Machine Learning & Uncertainty Quantification — *Flagstaff, AZ*

- Built multi-fidelity PyTorch surrogates predicting natural frequencies at reduced compute cost
- Trained a high-fidelity surrogate to **97.4% accuracy** in predicting bandgap locations
- Embedded physics-informed neural networks (PINNs) to boost accuracy on sparse training data

Project Engineering Intern — **Summer 2026**
Achen-Gardner Construction — *Phoenix, AZ*

- Drafted submittals, purchase orders, subcontracts, and RFIs for a $16M public-works project
- Cross-checked vendor submittals against materials lists, catching 10+ spec discrepancies
- Prepared Guaranteed Maximum Price (GMP) submittals for a City of Buckeye project

Projects

CivilFLOW — Retrieval-Augmented Document Engine | *Next.js, TypeScript, Claude API* | Live Site — **2026**

- Built a full-stack RAG engine drafting construction purchase orders, subcontracts, and RFIs
- Engineered hybrid BM25 + embedding retrieval, achieving **100% hit@5** and **1.000 MRR**
- Built an eval harness on a 14-case golden set tracking retrieval, citation, and schema checks
- Hardened outputs with JSON schemas, prompt-injection screening, and citation-grounding audits

GridCast — Probabilistic Electricity-Demand Forecasting | *Python, PyMC, scikit-learn* — **2026 – Present**

- Ingested **17,500+ hourly observations** of PJM electricity demand from the U.S. EIA API
- Designing calendar, lag, and holiday features plus a seasonal-naive walk-forward baseline
- Developing Kalman-filter and Bayesian time-series models (PyMC) for calibrated 90% intervals

Monte Carlo Options Pricer & Backtester | *Python, NumPy, pandas* | GitHub — **2026**

- Priced European options via 10,000-path Monte Carlo (GBM), validating against Black–Scholes
- Estimated realized volatility from AAPL log returns and plotted the implied-volatility smile
- Backtested a 21-month mean-reversion strategy (IV > 1.35x RV), surfacing short-gamma risk

Full-Stack Retirement Monte Carlo Simulator | *React, FastAPI, NumPy* | Live Site — **2026**

- Built a 2,000-path geometric Brownian motion simulation engine in Python and NumPy
- Computed median retirement timelines, success rates, and top/bottom-decile outcomes
- Deployed a React/Vite frontend (Vercel) with a FastAPI backend (Render) over a REST API

Math Lab — Putnam Competition Mathematics Database | *JavaScript, Python, KaTeX* | Live Site — **2026**

- Built a searchable database of 372 Putnam problems (1995–2025) as a dependency-free SPA
- Vendored KaTeX for offline math rendering over a schema-validated JSON data layer
- Automated verbatim extraction from archive TeX with a validator compiling every formula

Technical Skills

Languages: Python, SQL, C++, Java, JavaScript, TypeScript, MATLAB, LaTeX
ML & Data Science: PyTorch, scikit-learn, pandas, NumPy, PyMC, statsmodels, Bayesian inference
AI & Web: Claude API, RAG, React, Next.js, FastAPI, REST APIs, Git, Linux, Vercel